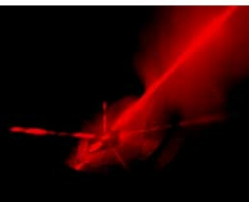


PLASMONICS PROMISES FASTER COMMUNICATION

Get ready to witness even faster computing and telecom. Plasmonics—a new technology—promises to bring this revolution by putting together the best of electronics and photonics



■ JAGMEET SINGH AND
DR S.S. VERMA

Currently, communication systems are based on either electronics or photonics. However, with the quest for transporting huge amounts of data at a high speed along with miniaturisation, both these technologies are facing limitations. Due to their mismatched capacities and sizes, it is very difficult to cobble them to get a high bitrate with miniaturisation.

So researchers are pioneering a new technology called 'plasmonics.' Due to its frequency being approximately equal to that of light and ability to interface with similar-size electronic components, plasmonics can act as a bridge between photonics and electronics for communication.

What is plasmonics?

The term 'plasmonics' is derived from plasmons—quanta associated with collective excitation of free electrons in metals. Surface plasmons are associated

with surface charge oscillations. Their frequency is almost equal to that of light; optical frequencies are about 10^5 times greater than the frequency of today's electronic microprocessors. So light can be used to excite them on the surface of a material in a localised regime.

The energy required to receive and send a surface plasmon pulse can be less than for electric charging of a metallic wire. This could allow plasmons to travel along nanoscale wires (called interconnects) carrying information from one part of a microprocessor to another with a high bitrate.

Plasmonic interconnects would be a great boon for chip designers, who have been able to develop ever smaller and faster transistors but have had a harder time building minute electronic circuits that can move data quickly across the chip.

Surface plasmons can be excited on a flat nano-film, nanostrip or other shaped nanoparticles such as nanosphere, nanorod, nanocube and nanostar. When nanoparticles are used to excite surface plasmons by light, these are known as localised surface plasmons. Silver and gold are of particular interest due to their high field enhancement and resonance wavelength lying in the visible spectral regime. The speed of these surface plasmons is almost equal to that of light with wavelength of the order of tens of nanometres.

Limitations of present modes

Presently, electronics plays an important role in communication. In labora-

tories, though, photonics has started replacing electronics where a high data transfer rate is required.

Electronics deals with the flow of charge (electrons). When the frequency of an electronic pulse increases, the electronic device becomes hot and wires become very loose. Hence by the principle of "the higher the frequency, the higher the data transfer rate," a huge amount of data cannot be transferred.

On the other hand, when the size of an electronic wire reduces, its resistance (inversely proportional to the cross-sectional area of the wire) increases but the capacitance remains almost the same. This leads to time delay effects.

In photonics, optical fibres (cylindrical dielectric/non-conducting waveguides) are used. These transmit light along their axis by the process of total internal reflection. The fibre consists of a core surrounded by a cladding layer, both of which are made of dielectric materials. To confine the optical signal in the core, the refractive index of the core must be greater than of the cladding. The lateral confinement size of the optical cable is approximately half the wavelength of the light used. Hence the size of the optical cable is of the order of hundreds of nanometres—larger than today's electronic devices.

Communication with plasmonics

Plasmonic structures can exert huge control over electromagnetic waves